COMP4141 Slides Week 10

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Admin

Sample Exam

- Has been released
- Solutions will be provided later
- Feel free to discuss with me :)

MyExperience

- I am sure you had an experience
- Please^{3000} fill it out!

Feedback From Marking

- Make sure to mention the reduction is polynomial time when trying to prove NP-hardness.
- Be clear on what your verifier is when you try to use one.
- Make sure to explain why your algorithm/verifier is polynomial time in NP inclusion proofs.

W9 P5

Problem 5

Consider the following decision problem

2-SAT

Input: A boolean formula φ in 2-CNF (at most 2 literals per clause)

Question: Is φ satisfiable?

- Last week I made a mistake; the condition for non-satisfiability is that there is a path from $\neg y$ to y and from y to $\neg y$ (i.e. bidirectional reachability).
- The same idea, using that NL is closed under complement, shows that we can find such paths for an unsatisfiable formula.
- If you know more graph theory, this bidirectional reachability means that y and $\neg y$ lie in the same strongly connected component (SCC).

PP

Probabilistic Algorithms

Probabilistic Algorithms

A probabilistic TM (PTM) M is a type of NTM in which each non-deterministic step is called a *coin-flip step* and has two legal next moves.
The probability of a branch b of M's computation on input w is

$$\Pr[b] = \frac{1}{2^k},$$

where k is the number of coin-flip steps that occur on b.
The acceptance probability is the sum of the probabilities of all accepting branches (the rejection probability is its complement).

PP

PP

Say that a PTM M probabilistically recognises A if it runs in polynomial time and:

1. $w \in A \implies \Pr[M \text{ accepts } w] > \frac{1}{2}$,
2. $w \notin A \implies \Pr[M \text{ rejects } w] > \frac{1}{2}$.

Define **PP** as the class of languages that are probabilistically recognised by polynomial-time PTMs.

Working with **PP**

For example, SAT $\in$ **PP**.

PP in the hierarchy

NP $\subseteq$ **PP** $\subseteq$ PSPACE

Bounded-error Probabilistic Polynomial Time

BPP

BPP

For $\varepsilon \in [0, \frac{1}{2})$, a PTM M recognises A with error probability ε if:

1. $w \in A \implies \Pr[M \text{ accepts } w] \geq 1 - \varepsilon$,
2. $w \notin A \implies \Pr[M \text{ rejects } w] \geq 1 - \varepsilon$.

BPP is the class of languages recognised by polynomial-time PTMs with error probability $\frac{1}{3}$.

BPP example

$\textsc{Primes} = \{n \mid n \text{ is prime (in binary)}\}$
$\textsc{Primes} \in \mathbf{BPP}$

Amplification Lemma

Amplification Lemma

Let $\varepsilon \in [0, \frac{1}{2})$ and let $p(n) > 0$ be a polynomial. If a polynomial-time PTM M recognises A with error probability ε, then there exists another polynomial-time PTM M' that also recognises A with error probability $\exp(-p(n))$.

Interpretation

The definition of **BPP** is robust with respect to the choice of error probability; $\frac{1}{3}$ is just one convenient choice in $[0, \frac{1}{2})$.

RP & ZPP

RP

RP

RP is the class of languages recognisable by probabilistic polynomial-time TMs such that:

- If $w \in A$, then $\Pr[M \text{ accepts } w] \geq \frac{1}{2}$,
- If $w \notin A$, then $\Pr[M \text{ rejects } w] = 1$.

(Also called Monte Carlo PTIME.)

$$\mathbf{coRP} = \{A \mid \overline{A} \in \mathbf{RP}\}, \quad \mathbf{ZPP} = \mathbf{RP} \cap \mathbf{coRP}.$$

Results in **RP**

Lemma

$\mathbf{RP} \subseteq \mathbf{BPP}$
$\mathbf{RP} \subseteq \mathbf{NP}$

Las Vegas Algorithms

Las Vegas

A probabilistic TM M is called *Las Vegas* if for all $w \in \Sigma^*$:

$$\Pr(M \text{ accepts } w) \in \{0, 1\},$$

and for some polynomial p, the expected running time of M on inputs of length n is at most $p(n)$.

Las Vegas and **ZPP**

ZPP is the class of languages recognised by Las Vegas algorithms.

ZPP Problem

ZPP characterisation

Show that **ZPP** can be characterised as the set of languages L such that there exists a probabilistic Turing machine which:

- Runs in polynomial time and halts in ACCEPT, REJECT, or DON'T KNOW,
- On input $x \in L$, halts in ACCEPT with probability $p > \frac{1}{2}$ or in DON'T KNOW,
- On input $x \notin L$, halts in REJECT with probability $p' > \frac{1}{2}$ or in DON'T KNOW.

ZPP Solution

If $L \in \mathbf{ZPP} = \mathbf{RP} \cap \mathbf{coRP}$, do the following on input w:

- Run w on the **RP** machine. If it accepts, then ACCEPT,
- Otherwise run w on the **coRP** machine. If it rejects, then REJECT,
- Otherwise output DON'T KNOW.

Conversely, suppose we have such a machine:

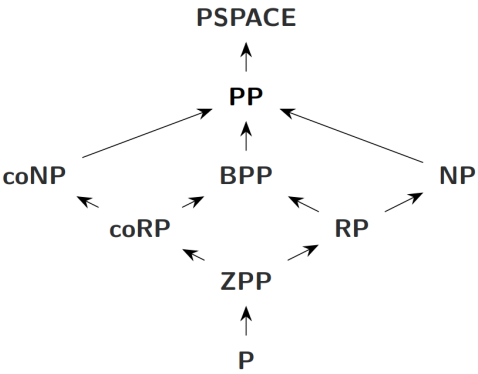
- If we output REJECT on DON'T KNOW, we obtain an **RP** algorithm,
- If we output ACCEPT on DON'T KNOW, we obtain a **coRP** algorithm.

Alternatively, to obtain a Las Vegas algorithm:

- On output DON'T KNOW, rerun the machine.

This yields finite expected running time.

Relationships



W10 P1

Problem 1

Probabilistic computers use a fair coin-flip to make probabilistic choices, is this enough to cover any probability distribution?

(a) Show how a fair coin can simulate rolling a dice.

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Probabilistic computers use a fair coin-flip to make probabilistic choices, is this enough to cover any probability distribution?

(b) Show how a fair coin can simulate making a choice with probability p for any $p \in (0, 1)$.

W10 P2

Problem 2

In the lecture, we showed that SAT $\in$ **PP** and concluded that **NP** $\subseteq$ **PP**. There is actually a missing step in the argument, relating to how **PP** interacts with polynomial time reductions. Formulate and prove a result about this, and use it to complete the argument.

W10 P3

Problem 3

Prove or disprove: **BPP** is closed under intersection. That is, if $A \in \mathbf{BPP}$ and $B \in \mathbf{BPP}$ then $A \cap B \in \mathbf{BPP}$.

W8 P5

Problem 5

Consider the following problem:

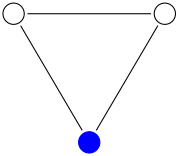
3-Col

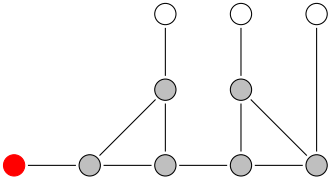
Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Problem 5

(a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?





Problem 5

(b) Show that 3-Col is **NP**-complete, assuming that SAT is **NP**-complete.

W8 P6

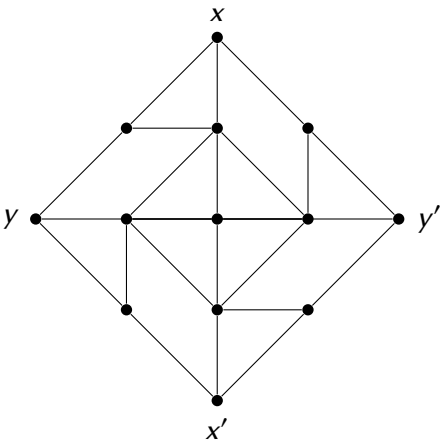
Problem 6

Show that if $\mathbf{P} = \mathbf{NP}$, then every language $A \in \mathbf{P}$ except $A = \emptyset$ and $A = \Sigma^*$, is $\mathbf{NP}$-complete.

W8 Bonus

Bonus Week 8

Consider the following graph:



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(a) Give a valid 3-colouring of this graph where x and y have different colours.

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(c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

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(c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

(d) Consider the following decision problem:

Planar 3-Col

Input: A planar graph G

Question: Does G have a valid 3-colouring?

Show that Planar 3-Col is NP-complete.
Hint: Use the above graph to "uncross" edges.

Bonus Week 8

(e) Prove that there is no planar graph G, with x, y, x', y' on an outer face (in that order), that satisfies the conditions indicated in (a), (b) and (c), but with a 4-colouring rather than a 3-colouring.